Yang Huang

Machine Learning / Operations Research Algorithm Intern

Institute of Automation, Chinese Academy of Sciences, Beijing

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Education

Institute of Automation, Chinese Academy of Sciences 2022–Present
Ph.D. Candidate, direct Ph.D. track Beijing

Research focus: graph neural networks and reinforcement learning for combinatorial optimization (ML4CO).

University of Science and Technology Beijing 2018–2022
B.S. in Mathematics and Applied Mathematics Beijing

Mathematics background; recipient of Outstanding Student Scholarships in multiple academic years.

Publications

ICML 2026 **First Author**. Optimal Transport–Guided Stochastic Control for Graph Combinatorial Optimization.
International Conference on Machine Learning (ICML 2026), Regular

TMLR **First Author**. Learning to Destroy and Repair Local Optima in Multilinear Graph QUBO Optimization.
Transactions on Machine Learning Research (TMLR), Under Review

Research Experience

OT-Guided Diffusion and Stochastic Control for Graph Combinatorial Optimization Jan. 2025–Sep. 2025
First Author / Lead Researcher Accepted by ICML 2026; Code

- Unified MIS, MaxCut and MaxClique as QUBO problems under a multilinear relaxation on $[0,1]^n$; proved no optimality gap and the existence of binary optimal solutions.
- Improved diffusion-style neural combinatorial optimization by replacing hand-designed or score-only sampling dynamics with an OT-guided controlled diffusion toward $\pi^\star(x) \propto \exp(-f(x;G))$.
- Derived the optimal drift from the entropic OT / Schrödinger bridge formulation and made it tractable via Laplace approximation, yielding a proximal energy-minimization target amortized by a GNN controller.
- Trained the controller from rollout-buffer states with Laplace-energy supervision; performed multi-particle Euler–Maruyama sampling and greedy decoding for feasible discrete solutions.
- Achieved average MIS sizes of 20.12/43.41 on RB-small/ER-large, improving over the best learning baseline by 0.50/0.07; achieved MaxCut values of 731.01/2965.94 on BA-small/BA-large, improving over DiffUCO by 3.69/18.41.

Learned Destroy-and-Repair Test-Time Search for Multilinear Graph QUBO Oct. 2025–Present
First Author / Core Algorithm Designer Submitted to TMLR; Code

- Studied nonconvex local optima in multilinear QUBO relaxations, where direct gradient optimization is slow and amortized neural solvers can repeatedly converge to suboptimal binary solutions.
- Designed a recurrent repair network as a learned energy-based optimizer, minimizing the original relaxed QUBO energy from random or damaged continuous states before discrete decoding.
- Designed a destroy policy for test-time refinement: it observes graph structure, current state and search step, then converts node-level probabilities into a learned destruction rate for randomized masking.
- Trained repair and destroy alternately using random damage, policy-gradient rewards from decoded-score improvement, and repair fine-tuning over full destroy-repair trajectories.
- With the same 512 repair-call budget, improved over repair-only by 2.16/5.91 MIS vertices and 2.06/11.32 MaxCut value on small/large graphs.

Skills

Programming	Python, PyTorch, DGL, NumPy, NetworkX; GPU training, batched evaluation and experiment logging
ML / RL	Graph neural networks, policy gradient, stochastic differential equation sampling, multi-particle inference
Optimization	QUBO, multilinear relaxation, optimal transport / stochastic control, local search, MIS / MaxCut / MaxClique
Tools	Linux, tmux, Git, LaTeX; ablation design and paper reproduction